```
import numpy as np
import pandas as pd
import torch
import umap
import hdbscan
from collections import defaultdict
import json
import matplotlib.pyplot as plt
```

```
PALABRAS_CSV = {  # salida de reconstruir_palabras.py, por modelo
    "mex_state":   "output/words_mex_state.csv",
    "mex_large":   "output/words_mex_large.csv",
    "beto":        "output/words_beto.csv",
    "bertin":      "output/words_bertin.csv",
    "robertuito":  "output/words_robertuito.csv",
}
```

```
def get_tweet(palabras_df, id_tweet, special_token=None):

    RED = "\033[91m"
    RESET = "\033[0m"


    l = list(palabras_df[palabras_df['id_tweet']==id_tweet]['word_normalized'])
    l = [str(_) for _ in l]

    if special_token==None:
        print(" ".join(l))
        return

    s = ""
    for w in l:
        if w==special_token:
            s += RED + w + RESET + " "
        else:
            s += w + " "
    print(s)
    return
```

```
model_name = 'beto'
```

```
def cargar_embeddings(nombre_modelo, carpeta_salida='embeddings'):
    data = np.load(f"embeddings/{nombre_modelo}_embeddings.npz", allow_pickle=True)
    X, words, id_t = data["embeddings"], data["words"], data['id_tweets']
    return X, words, id_t

data = pd.read_csv(PALABRAS_CSV[model_name])

emb, w, id_tweets = cargar_embeddings(model_name)
emb /= np.linalg.norm(emb, axis=1).reshape(-1, 1)
df = pd.DataFrame(list(w), columns=['palabra'])
df['id_tweet'] = id_tweets
perm = df['id_tweet'].argsort().tolist()
df = df.iloc[perm]
emb = emb[perm]
# agregar la región a las palabras en df
region_por_tweet = data.drop_duplicates("id_tweet").set_index("id_tweet")["region"]
df["region"] = df["id_tweet"].map(region_por_tweet)
df['region_codes'] = pd.Categorical(df['region']).codes
```

```
df.head(2)
```

	palabra	id_tweet	region
0	lava	0	Jalisco
4247	salga	0	Jalisco

```
df = df[0:100000]
```

```
reducer = umap.UMAP(metric='cosine', n_components=2, n_neighbors=15)
u = reducer.fit_transform(emb[0:100000])
```

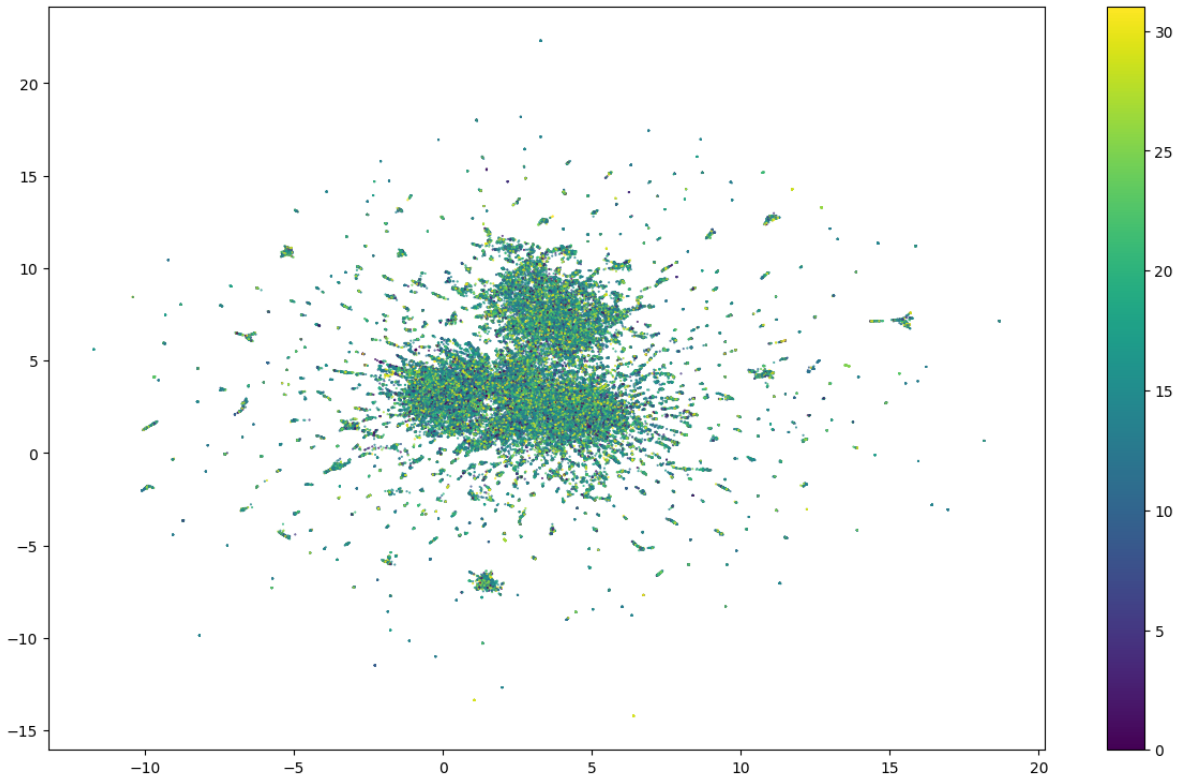
```
C:\Users\msubr\Anaconda3\envs\gpu_env\lib\site-packages\umap\spectral.py:548: UserWarning: Sp
failed. This is likely due to too small an eigengap. Consider
adding some noise or jitter to your data.

Falling back to random initialisation!
  warn(
C:\Users\msubr\Anaconda3\envs\gpu_env\lib\site-packages\umap\spectral.py:548: UserWarning: Sp
failed. This is likely due to too small an eigengap. Consider
```

```
adding some noise or jitter to your data.

Falling back to random initialisation!
  warn(
```

```
plt.figure(figsize=(15, 9))
scatter = plt.scatter(u[:, 0], u[:, 1], s=.2, c=df['region_codes'])
plt.colorbar(scatter, label="")
plt.show()
```




```
from sklearn.cluster import DBSCAN
```

```
clr = DBSCAN(eps=.1, min_samples=8)
cluster_labels = clr.fit_predict(u)
```

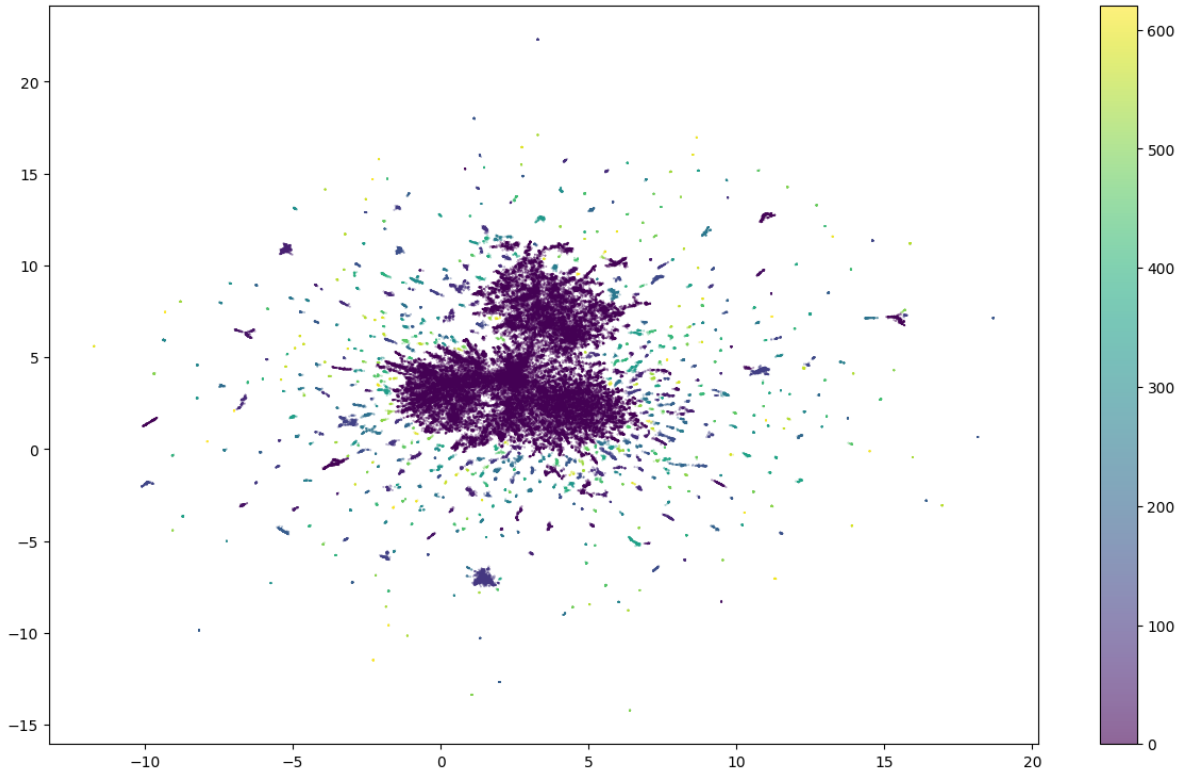
```
df['dbscan_cluster'] = cluster_labels
```

```
cluster, count = np.unique(cluster_labels, return_counts=True)
len(cluster)
```

622

```
s = u[cluster_labels > -1]
cl = cluster_labels[cluster_labels > -1]
print("Ruido:", len(u) - len(s))
plt.figure(figsize=(15, 9))
scatter = plt.scatter(s[:, 0], s[:, 1], s=.1, c=cl, alpha=0.6, )
plt.colorbar(scatter, label="")
plt.show()
```

Ruido: 875



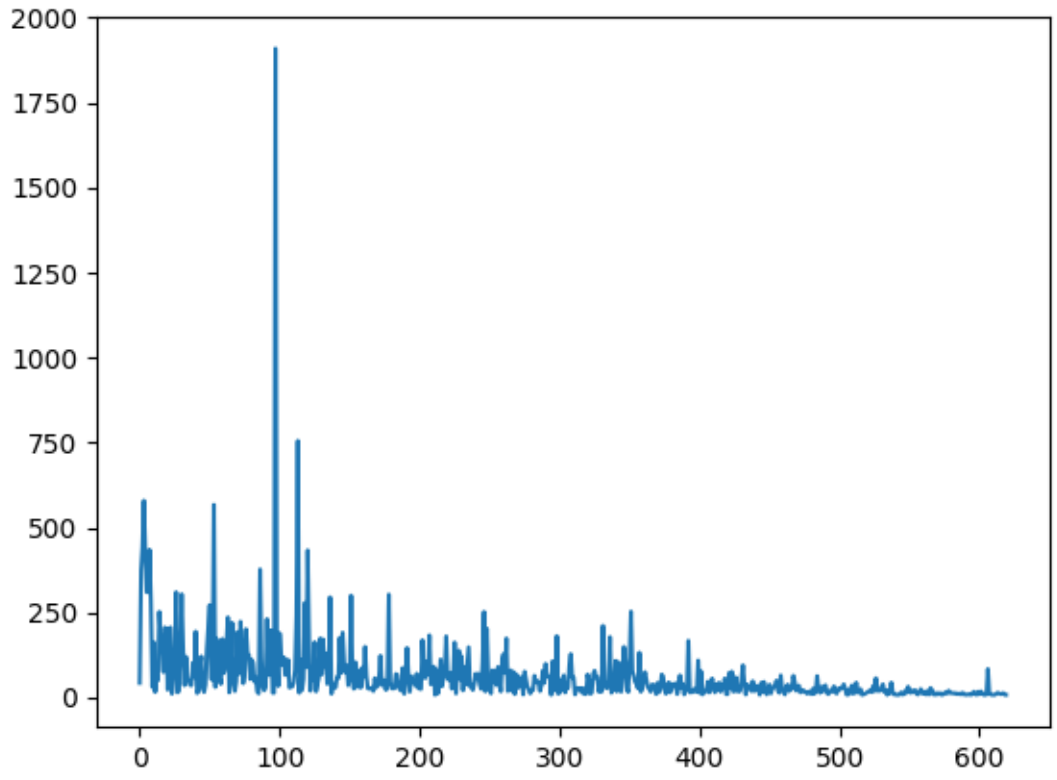

```
tokens_in_clusters = (df
                      .groupby("palabra")['dbscan_cluster']
                      .unique()
```

```
                        .reset_index(name="clusters")
                       )
df2 = df.groupby("palabra")['dbscan_cluster'].nunique().reset_index(name="num_clusters")
tokens_in_clusters['num_clusters'] = df2['num_clusters']
```

```
tokens_in_clusters.head(2)
```

	palabra	clusters	num_clusters
0	aaaaah	[0]	1
1	aaaah	[0, 202]	2

```
plt.plot(count[2:])
plt.show()
```




```
mask = count < 1000
decent_clusters = []
```

```
num_decent_clusters = []
for i, d in tokens_in_clusters.iterrows():
    c = d['clusters']
    c_ = []
    for j in range(len(c)):
        if mask[c[j]+1]:
            c_.append(c[j])
    decent_clusters.append(c_)
    num_decent_clusters.append(len(c_))
```

```
tokens_in_clusters['decent_clusters'] = decent_clusters
tokens_in_clusters['num_decent_clusters'] = num_decent_clusters
```

```
tokens_in_clusters['num_decent_clusters'].unique()
```

```
array([0, 1, 2, 6, 3, 4, 5])
```

```
print(tokens_in_clusters.shape)
tokens_in_clusters.head(2)
```

```
(11333, 5)
```

	palabra	clusters	num_clusters	decent_clusters	num_decent_clusters
0	aaaaah	[0]	1	[]	0
1	aaaah	[0, 202]	2	[202]	1

```
same_number_clusters = tokens_in_clusters[tokens_in_clusters['num_decent_clusters'] == 3]
same_number_clusters_tokens = []
for i, d in same_number_clusters.iterrows():
    #if (len(d['palabra']) < 4):
    #    continue
    same_number_clusters_tokens.append(d['palabra'])
best_tokens = []
for token in same_number_clusters['palabra']:
    token_clusters = tokens_in_clusters[tokens_in_clusters['palabra']==token]['decent_cluste
    token_clusters = token_clusters.to_list()[0]

    for tk_cl in token_clusters:
```

```
        df_ = df[df['dbscan_cluster'] == tk_cl]
        id_tweets = df_[df_['palabra']==token]['id_tweet'].to_list()
        if (len(id_tweets) < 5):
            break
    else:
        best_tokens.append(token)
```

```
len(best_tokens), list(enumerate(best_tokens))
```

```
(8,
 [(0, np.str_('aguirre')),
  (1, np.str_('cambio')),
  (2, np.str_('charlar')),
  (3, np.str_('clase')),
  (4, np.str_('domingo')),
  (5, np.str_('horas')),
  (6, np.str_('medio')),
  (7, np.str_('siento'))])
```

```
token = best_tokens[1]

token_clusters = tokens_in_clusters[tokens_in_clusters['palabra']==token]['decent_clusters']
token_clusters = token_clusters.to_list()[0]
print(token)
#print(cluster_dist(token_clusters[0], token_clusters[1], s, cl))
for tk_cl in token_clusters:
    print("_"*50)
    df_ = df[df['dbscan_cluster']==tk_cl]
    print(tk_cl, len(df_))

    id_tweets = df_[df_['palabra']==token]['id_tweet'].to_list()
    print("Total tweets:", len(id_tweets))
    print("Tokens in cluster:", df_['palabra'].unique())
    for id_t in id_tweets[0:5]:
        print()
        get_tweet(data, id_t, special_token=token)
```

```
cambio
--------------------------------------------------
354 64
```

```
Total tweets: 38
Tokens in cluster: [np.str_('cambio') np.str_('modificaciones') np.str_('cambios')
 np.str_('giro') np.str_('ajustes') np.str_('transformaciones')]

_ usr a esta señora yo creo que lo que le hace falta es que lleve su carro a que le hagan el

no podemos cambiar no podemos alejarnos de lo que somos hasta que aceptamos lo que somos ento

_ usr así se verán los hoteles hilton ahora que implementen la medida de solo asear los cuart

y porqué lo siguen haciendo no que representaban un cambio qué pena que se ma luchadora como

agradece la gota que colmo el vaso es la semilla del cambio que pedias en ciudad de mexico co
--------------------------------------------------
150 36
Total tweets: 11
Tokens in cluster: [np.str_('efecto') np.str_('realidad') np.str_('cambio')]

exactamente se trata de parar los moches o regalias a cambio de votos si no se aprueba ahora

el encanto del amor consiste también en una especie de permanente amenaza en cambio la amista

no me llores porque si lloras yo peno en cambio si tú me cantas yo siempre vivo y nunca muero

la prostitución en el mundo trans es mucho más que un trabajo ser inevitablemente parte de un

el trabajo sexual para el mundo trans es mucho más que un trabajo ganar dinero a cambio del c
--------------------------------------------------
368 38
Total tweets: 10
Tokens in cluster: [np.str_('mundial') np.str_('cambio') np.str_('global')
 np.str_('calentamiento')]

atentar contra la sal ningún país ni empresa va a querer invertir en estás condiciones va a a

pide diputada ejercer la ley del cambio climático _ url _ usr

_ usr una solución efectiva es aumentar el consumo de frijoles orgánicos veganos sin gluten l

lopéz obrador entiende bien a la gran mayoría de los mexicanos aprovecha esos resentimientos

laresumidera de lainformadera anuncian operativo para romería 2021 onu premia al amg por sus
```